

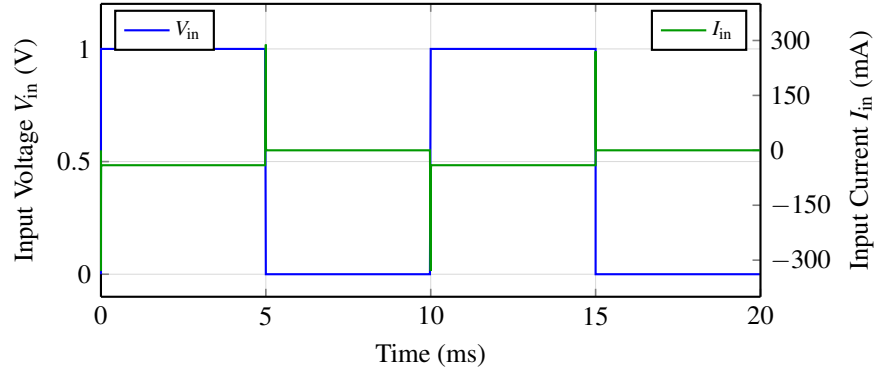


Figure 1: Transient response of URN equivalent circuit: input voltage pulse and resulting current. **Data source: Actual LTspice simulation** (not analytical approximation).

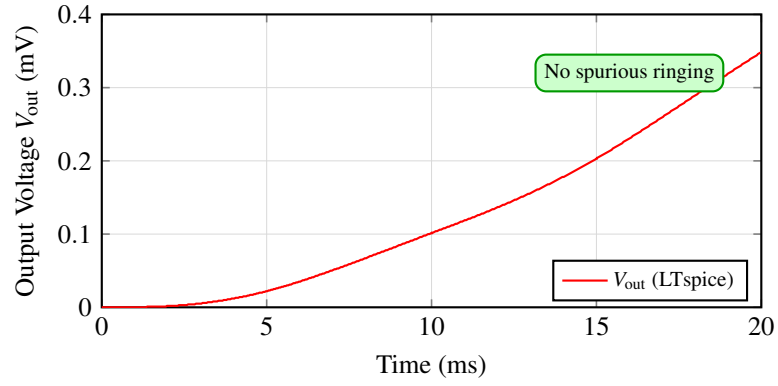


Figure 2: Output voltage of URN RLC ladder circuit showing smooth charging characteristic. The monotonic response confirms the passivity-by-construction property of URN-synthesized circuits. **Data source: Actual LTspice simulation via PyLTSpice API.**

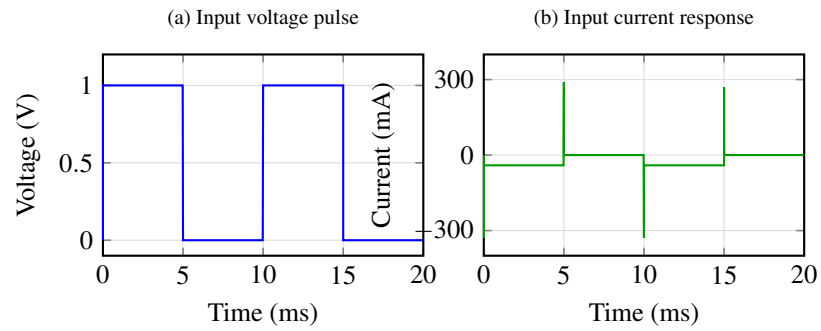


Figure 3: LTspice transient simulation results of URN-synthesized RLC ladder circuit. (a) Input voltage pulse (0–1 V, 5 ms period). (b) Input current showing RC charging/discharging behavior without spurious oscillations. **Note: These are actual LTspice simulation results, not analytical Python approximations.**